Erratum to Eq. (4):

$$I(\tau_i, \mu_\pm, \varphi) = I(\tau_{i\mp1}, \mu_\pm, \varphi)\exp\left(\mp\frac{\Delta\tau}{\mu_\pm}\right) + I_1(\Delta\tau, \mu_\pm, \varphi) \pm \frac{1}{\mu_\pm}\int_{\Delta\tau} J(t, \mu_\pm, \varphi)\exp\left(\mp\frac{t}{\mu_\pm}\right)dt \quad (4)$$

In Eq. (4), the terms have the following physical meaning (Fig. 4): intensity at some level i (left hand side) is the sum of the intensity at the neighboring level above $i-1$ (for descending light, $\mu_+ > 0$) or below $i+1$ (for ascending light, $\mu_- < 0$) the current level, plus single scattering I_1 within the level $\Delta\tau$ (solar beam attenuated appropriately for embedded element layer $\Delta\tau$), plus multiple scattering

In Eq.(4), the right-most term must be different for upward and downward radiation.

A) For upward radiation, Eq.(4) is correct: exponential attenuation occurs between the level *t* and the top of the layer element Δτ, where *t* = 0 (see Fig. 4)

$$-\frac{1}{\mu_-}\int_{\Delta\tau} J(t, \mu_-, \varphi)\exp\left(+\frac{t}{\mu_-}\right)dt \approx -\bar{J}(\mu_-, \varphi)\int_{\Delta\tau}\exp\left(\frac{t}{\mu_-}\right)d\left(\frac{t}{\mu_-}\right) =$$

$$= -\bar{J}(\mu_-, \varphi)\exp\left(\frac{t}{\mu_-}\right)\Bigg|_0^{\Delta\tau} = \bar{J}(\mu_-, \varphi)\exp\left(\frac{t}{\mu_-}\right)\Bigg|_{\Delta\tau}^{0} =$$

$$= \bar{J}(\mu_-, \varphi)\left[1-\exp\left(\frac{\Delta\tau}{\mu_-}\right)\right]$$

B) For downward radiation, exponential attenuation occurs along the "residual" path, $\Delta\tau - t$:

$$+\frac{1}{\mu_+}\int_{\Delta\tau} J(t, \mu_+, \varphi)\exp\left(-\frac{\Delta\tau - t}{\mu_+}\right)dt \approx \bar{J}(\mu_+, \varphi)\int_{\Delta\tau}\exp\left(-\frac{\Delta\tau - t}{\mu_+}\right)d\left(\frac{t}{\mu_+}\right) =$$

$$= \bar{J}(\mu_+, \varphi)\exp\left(-\frac{\Delta\tau}{\mu_+}\right)\int_{\Delta\tau}\exp\left(\frac{t}{\mu_+}\right)d\left(\frac{t}{\mu_+}\right) = \bar{J}(\mu_+, \varphi)\exp\left(-\frac{\Delta\tau}{\mu_+}\right)\exp\left(\frac{t}{\mu_+}\right)\Bigg|_0^{\Delta\tau} =$$

$$= \bar{J}(\mu_+, \varphi)\exp\left(-\frac{\Delta\tau}{\mu_+}\right)\left[\exp\left(\frac{\Delta\tau}{\mu_+}\right)-1\right] = \bar{J}(\mu_+, \varphi)\left[1-\exp\left(-\frac{\Delta\tau}{\mu_+}\right)\right]$$

Compare with (A) above and see Eq.(37) in the paper.

Fig.4 also shows the right-most element from Eq.(4) inaccurately.

The text after Eq. (4) – see at the top of the page – is also missing the fact that intensity from the neighboring levels contribute to the current one after *exponential attenuation*, which is present in Eq. (4): see the first term to the right of the equality sign.

These typos do not affect the correctness of the GSIT code itself, but they affect the proper understanding of the details. (S. Korkin, 2026/10/12)